

CS 791

Probabilistic Foundations of AI

Course Overview

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CS 791 is renaming of CS 726: Advanced Machine Learning

Why rename?

- CS 726 was started 15 years back when ML was a niche field of CS, and the department ran two courses: Foundation of ML and Advanced ML.
 - Topics evolved every year as a Markov process but the focus on probabilities has persisted.
- The field has grown 10-100 fold now along many dimensions:
 - number of topics, number of researchers, number of papers, amount of funding, number of AI/ML courses in CSE (2→23!)
- Want a name that more faithfully respects the true contents of the course, and IITB does not allow just title change.



Scope of the course

Advanced topics in probabilistic modeling that forms the foundations of modern AI.

Learning to represent, train, generate, and reason on object :

- High dimensional $x = \{x_1, \dots, x_n\}$, space of x is large
- Inter-dependent components

Examples: Image, Video, Time-series, Text

Examples of high dimensional spaces



This image is very high-dimensional: comprising of $1024 \times 1024 \times 3 = 3 \text{ million}$ dimensional real space

Words in a sentence

If you ask a question, you are a fool only once. If you do not ask, you are a fool forever.

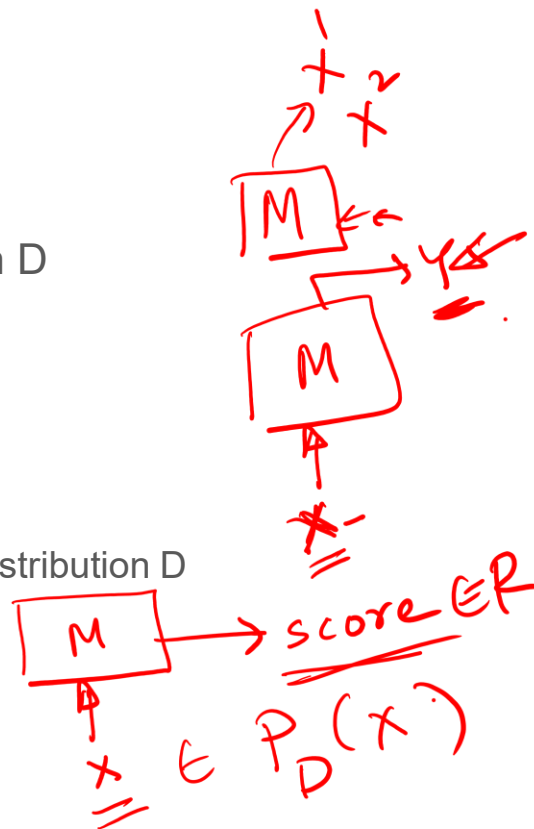
Assume a vocabulary size of 50 K.

The sentence of 25 words has $25 \times 50 \text{ K} = 1.25 \text{ million}$ dimensional discrete space

Different task settings

Given training data D , train a model M that can be used for

- Generation
 - Unconditional: Generate a sample X that is representative in D
 - Conditional: Given an input prompt X , generate a sample Y .
- Density estimation:
 - What is the probability that a given sample X is part of the training distribution D
- Other forms of reasoning:
 - Causality, Counter-factual reasoning, recourse on predictions.



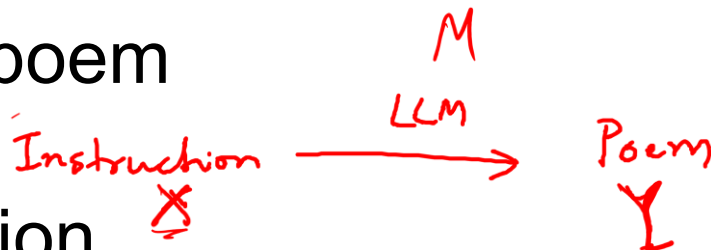
Importance of Probabilistic Modeling

A common task: Predict or generate an output in response to an input

- We may have insufficient data about the task to learn to predict perfectly
- The task itself may be stochastic: There is no one correct translation for a sentence, but some translations are more typical than others.
- If models are treated as probability distributions: we can attach confidence with generations.
- Rich theory allows for principled ways of decomposing high-dimensional modeling tasks into modular parts.

Text to text generation

- Write a poem



- Translation

- Text-to-tree generation

Translation

Input: $\mathbf{x}$

Predicted sequence: $\mathbf{y}$

Where can I find healthy and traditional Indian food?

→ स्वस्थ और पारंपरिक भारतीय भोजन कहाँ मिल सकता है?



- Each token in the output is a random variable and there is inter-dependence in the output tokens.
- We want to output a probability with the output translation, and not just produce one translation.
- We cannot predict the whole sentence in one shot but need to decompose it into parts

Text to image generation

- Imagen
- Stable diffusion

Topics for Generation

Goal: Output a distribution $P_{\theta}(\mathbf{y}|x)$ over a structured output $\mathbf{y} = y_1, \dots, y_n$, optionally conditioned on an input x .

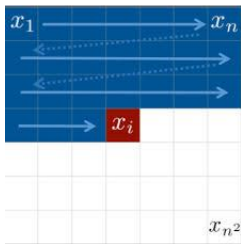
- **Representation/Modeling**: Form of P_{θ} , how to represent $P(\mathbf{y})$ of high-dimensional $\mathbf{y}$ for easy learnability and efficient inference.
- **Training or learning**: How to parameterize the distribution and learn the parameters
- **Inference**: How to efficiently generate?

Key insight from the course

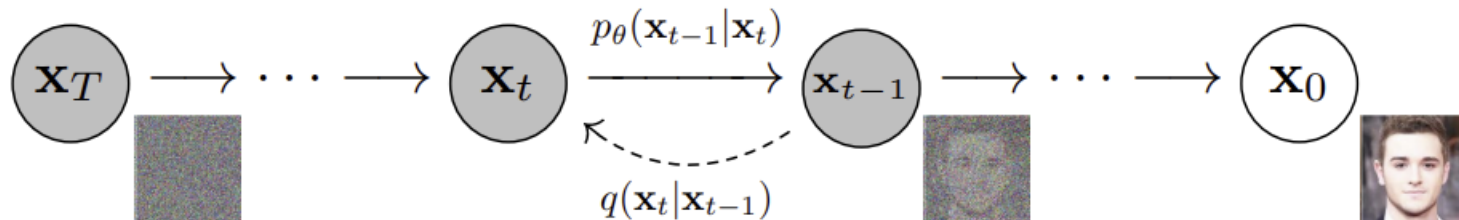
Decompose high-dimensional objects into smaller manageable sub-parts

Representation

- With observed variables



- With latent variables



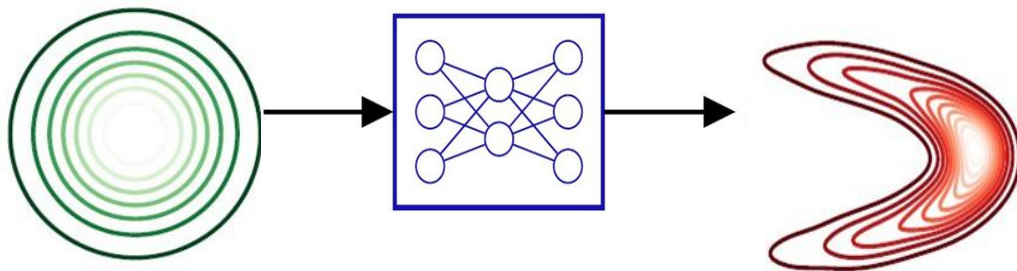
Can we make the dependency graph simpler via factorization?

Representation

- Represent the rate of change of a random variable (stochastic differential equations)

$$\frac{dX(t)}{dt} = f(X(t), t) + \sigma(X(t), t)W_t \quad W_t \sim \mathcal{N}(0, \mathbb{I})$$

- Represent as an invertible transformation of a simpler distribution



Learning

- How to parameterize the joint distribution for sample-efficient learning
- How to efficiently learn the parameters θ of the distribution
 - Training data (conditional): $D = \{(x^1, \mathbf{y}^1), \dots, (x^N, \mathbf{y}^N)\}$
 - Training data: (unconditional) $D = \{x^1, x^2, \dots, x^N\}$

Adapting trained distributions

- In-context learning for regression, time-series, and language tasks
- Parameter efficient fine-tuning

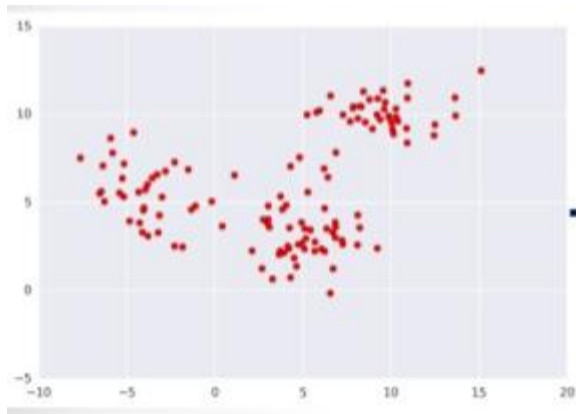
Inference

- Given an x , how to efficiently find the most likely $y_1, \dots, y_n$: MAP Inference.
- How to generate multiple representative examples from estimated model: Sampling
 - Generate examples that are representative of the distribution

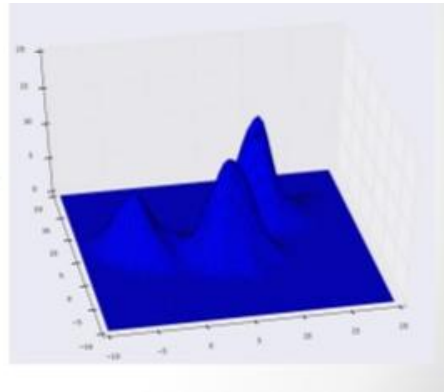
Density estimation

Given $D = \{x^1, x^2, \dots, x^N\}$ learn a $P(x)$, so that given a new x we can efficiently calculate the probability of “ x ”.

Applications: Out of distribution detection, outlier detection, classification



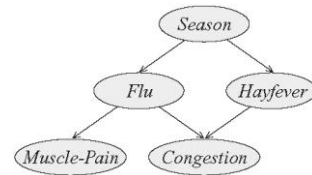
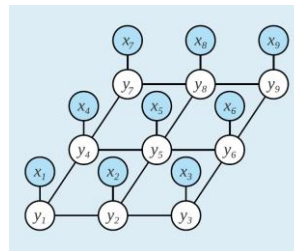
Density estimator



Course contents

Representation of $P(X)$ or $P(Y|X)$

- Probabilistic graphical models: Bayesian Networks and Markov Random Fields
 - Exact, efficient, but limited capacity
 - But, important to understand them to build a framework for probabilistic reasoning
 - Intuitive and easy to incorporate prior knowledge and biases
 - Special Graphical models
 - Gaussian processes: special structure that allow trivial computation of marginals



Representation (continued)

- Deep latent variable models:
 - VAEs, GANs, Discrete diffusion models – technology behind latest image generation models such as ImageGen
- Representation via variable transformation: Normalizing flows
- Stochastic differential equations $P(Y|X)$ where X is time and distribution represented as rate of change → continuous time diffusion model

Course contents

Learning

- Parameterization (model architectures for efficient learning)
 - Feature-based like in CRFs
 - Deep neural methods e.g. transformers
- Training algorithms
 - Maximum likelihood learning
 - Generalized Expectation Maximization: Variational Auto Encoders, diffusion models for images

Learning (continued)

- Advanced topics from deep learning:
 - In-context learning in foundation models
 - Parameter efficient fine-tuning
 - Model editing

Course contents

Inference

- Boolean queries on conditional inference
- Marginalization queries: $P(X_i)$, $\max_x P(x)$
 - Sum-product and max-product Inference in Graphical Models
- Sampling
 - Classical methods of sampling in tractable model: forward sampling, importance weighted sampling, Markov Chain Monte Carlo sampling (MCMC), Sequential Monte-Carlo
 - Recent methods usable in deep learning: Monte-Carlo with Langevin dynamics

Inference (Continued)

- Inference challenges in modern LLMs (a special Bayesian network)
 - Limitations of greedy decoding
 - Sampling multiple generations
 - Grammar constrained decoding
 - Speculative decoding
- Other forms of Inference
 - Causal effects
 - Algorithmic recourse

Who should take the course

- Students who are interested in doing research in machine learning
- Students who want to learn to think about learning from a probabilistic perspective in the context of modern deep learning
- Students who want to model learning tasks in a manner that cuts across applications.
 - The course will cite applications in NLP, vision, time-series, event sequences, and speech when relevant but it is not primarily about any of these applications.

Mode of running the course

- Two 85 minute slots per week:
- SAFE/Moodle quiz on the material covered in the **prior** week
 - 20 minute duration at a pre-announced time.
 - Grading will be done on top $n-2$ out of n quizzes. No compensation for missed quizzes.
- First mandatory quiz on July 31st on probability, linear algebra, optimization, and ML basics (The marks for this quiz will just be used to threshold)
- All materials will be uploaded on Moodle, announcements via Moodle, questions on Moodle or cs791@googlegroups.com
 - Forum for each topic for discussions and questions.

Evaluation

Approximate credit structure

- 15% In-class Quizzes
- 22% 4—6 graded programming and paper homeworks (in teams of 3)
- 25% Mid-semester exam
- 35% End semester exam
- 3% Editing scribe notes of previous batch, Attendance and class participation

Course calendar <https://www.cse.iitb.ac.in/~sunita/cs791/>